

L5 – 3.4 Solve Rational Equations and Inequalities

MHF4U

Jensen

Part 1: Rational Expressions

Rational Expression: the quotient of two polynomials, $\frac{p(x)}{q(x)}$, where $q(x) \neq 0$

Example 1: Simplify and state the restrictions of each rational expression

a) $\frac{2x^2-8}{x^2+3x+2}$

$$\begin{aligned} & \frac{2x^2-8}{x^2+3x+2} \\ &= \frac{2(x^2-4)}{(x+2)(x+1)} \quad \leftarrow \text{DOS} \\ &= \frac{2(x-2)(\cancel{x+2})}{(\cancel{x+2})(x+1)} \\ &= \frac{2(x-2)}{x+1} ; x \neq -2, -1 \end{aligned}$$

b) $\frac{x^3-x^2-x+1}{3x^3-3}$

$$\begin{aligned} & \frac{x^3-x^2-x+1}{3x^3-3} \\ &= \frac{(x^3-x^2)+(-x+1)}{3(x^3-1)} \quad \leftarrow \text{DOS} \\ &= \frac{x^2(x-1)-1(x-1)}{3(x-1)(x^2+x+1)} \\ &= \frac{(\cancel{x-1})(x^2-1)}{3(\cancel{x-1})(x^2+x+1)} \quad \leftarrow \text{DOS} \\ &= \frac{(x-1)(x+1)}{3(x^2+x+1)} ; x \neq 1 \end{aligned}$$

Part 2: Solve Rational Equations

Steps for solving rational equations:

- 1) Fully factor both sides of the equation
- 2) Multiply both sides by a common denominator (cross multiply if appropriate)
- 3) Continue to solve as you would a normal polynomial equation
- 4) State restrictions throughout (values of x that would make denominator equal zero)

Example 2: Solve each equation

a) $\frac{4}{3x-5} = 4$

$$\frac{4}{3x-5} = 4$$

$$(3x-5) \left(\frac{4}{3x-5} \right) = 4(3x-5) ; x \neq \frac{5}{3}$$

$$4 = 4(3x-5)$$

$$4 = 12x - 20$$

$$24 = 12x$$

$$\boxed{x = 2}$$

b) $\frac{6}{x-2} = x - 1$

$$\frac{6}{x-2} = x - 1$$

$$(x-2) \left(\frac{6}{x-2} \right) = (x-2)(x-1) ; x \neq 2$$

$$6 = x^2 - 1x - 2x + 2$$

$$0 = x^2 - 3x - 4$$

$$0 = (x-4)(x+1)$$

$$\swarrow$$

$$x-4=0$$

$$\boxed{x_1 = 4}$$

$$\searrow$$

$$x+1=0$$

$$\boxed{x_2 = -1}$$

c) $\frac{x-5}{x^2-3x-4} = \frac{3x+2}{x^2-1}$

$$\frac{x-5}{x^2-3x-4} = \frac{3x+2}{x^2-1}$$

$$\frac{x-5}{(x-4)(x+1)} = \frac{3x+2}{(x-1)(x+1)} ; x \neq -1, 1, 4$$

$$\frac{(x-1)(x+1)(x-5)}{x+1} = \frac{(x-4)(x+1)(3x+2)}{x+1}$$

$$(x-1)(x-5) = (x-4)(3x+2)$$

$$x^2 - 6x + 5 = 3x^2 - 10x - 8$$

$$0 = 2x^2 - 4x - 13 \quad \checkmark \text{ NOT Factorable use QF}$$

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(2)(-13)}}{2(2)}$$

$$x = \frac{4 \pm \sqrt{120}}{4}$$

$$x = \frac{4 \pm 2\sqrt{30}}{4}$$

$$x = \frac{2(2 \pm \sqrt{30})}{4}$$

$$x = \frac{2 \pm \sqrt{30}}{2}$$

Part 3: Solve Rational Inequalities

REMEMBER: Solving inequalities is the same as solving equations. However, when both sides of an inequality are multiplied or divided by a negative number, the inequality sign must be reversed.

Steps for solving rational inequalities algebraically:

- 1) Use inverse operations to move all terms to one side of the inequality
- 2) Combine the expressions by using a common denominator
- 3) Factor the polynomial
- 4) Find the interval(s) where the function is positive or negative by making a factor table

To make a factor table:

- Use x -intercepts and vertical asymptotes to divide in to intervals
- Use a test point within each interval to find the sign of each factor
- Determine the overall sign of the product by multiplying signs of each factor within each interval.

Example 3: Solve each inequality algebraically

a) $\frac{x^2+6x+5}{2x^2-7x+3} < 0$

$\frac{x^2+6x+5}{2x^2-7x+3} < 0$
 $\frac{(x+5)(x+1)}{(2x-1)(x-3)} < 0$
 Zeros: $x = -5, -1$
 Restrictions: $x \neq \frac{1}{2}, 3$

	$-\infty$	-6	-2	-1	0	$\frac{1}{2}$	1	3	4	∞
$x+5$		-	+	+	+	+	+	+	+	
$x+1$		-	-	+	+	+	+	+	+	
$2x-1$		-	-	-	-	+	+	+	+	
$x-3$		-	-	-	-	-	-	+	+	
overall		+	-	+	+	-	-	+	+	

The inequality is true when $-5 < x < -1$ OR $0.5 < x < 3$

The inequality is true when $x \in (-5, -1) \cup (0.5, 3)$

$$b) x - 2 < \frac{8}{x}$$

$$x - 2 < \frac{8}{x}$$

$$x - 2 - \frac{8}{x} < 0$$

$$\frac{x^2}{x} - \frac{2x}{x} - \frac{8}{x} < 0$$

$$\frac{x^2 - 2x - 8}{x} < 0$$

$$f(x) \left\{ \frac{(x-4)(x+2)}{x} < 0 \right. \begin{array}{l} \text{Zeros: } x = -2, 4 \\ \text{restriction: } x \neq 0 \end{array}$$

	$-\infty$	-2	0	4	∞
	-3	-1	1	5	
$x-4$	$-$	$-$	$-$	$+$	
$x+2$	$-$	$+$	$+$	$+$	
x	$-$	$-$	$+$	$+$	
overall	$(-)$	$+$	$(-)$	$+$	

The inequality is true when $x < -2$ OR $0 < x < 4$

The inequality is true when $x \in (-\infty, -2) \cup (0, 4)$

$$c) \frac{x+3}{x+1} \geq \frac{x-2}{x-3}$$

$$\frac{x+3}{x+1} \geq \frac{x-2}{x-3}$$

$$\frac{x+3}{x+1} - \frac{x-2}{x-3} \geq 0$$

$$\frac{(x+3)(x-3) - (x-2)(x+1)}{(x+1)(x-3)} \geq 0$$

$$\frac{x^2-9 - (x^2-x-2)}{(x+1)(x-3)} \geq 0$$

$$\frac{x^2-9 - x^2 + x + 2}{(x+1)(x-3)} \geq 0$$

$$\frac{x-7}{(x+1)(x-3)} \geq 0$$

zero: $x=7$
restrictions: $x \neq -1, 3$

	$-\infty$	-1	3	7	∞
		-2	0	4	8
$x-7$		$-$	$-$	$-$	$+$
$x+1$		$-$	$+$	$+$	$+$
$x-3$		$-$	$-$	$+$	$+$
overall		$-$	$(+)$	$-$	$(+)$

The inequality is true when $-1 < x < 3$ OR $x \geq 7$

The inequality is true when $x \in (-1, 3) \cup [7, \infty)$

$$d) \frac{x^3 + 6x^2 - 2x}{x^2 + 4} \geq 2$$

$$\frac{x^3 + 6x^2 - 2x}{x^2 + 4} \geq 2$$

$$\frac{x^3 + 6x^2 - 2x}{x^2 + 4} - 2 \geq 0$$

$$\frac{x^3 + 6x^2 - 2x}{x^2 + 4} - \frac{2(x^2 + 4)}{x^2 + 4} \geq 0$$

$$\frac{x^3 + 6x^2 - 2x - 2(x^2 + 4)}{x^2 + 4} \geq 0$$

$$\frac{x^3 + 6x^2 - 2x - 2x^2 - 8}{x^2 + 4} \geq 0$$

$$\frac{x^3 + 4x^2 - 2x - 8}{x^2 + 4} \geq 0$$

$$\frac{x^2(x+4) - 2(x+4)}{x^2 + 4} \geq 0$$

$$\frac{(x+4)(x^2-2)}{x^2+4} \geq 0$$

zeros: $x+4=0$
 $x=-4$

$x^2-2=0$
 $x^2=2$
 $x=\pm\sqrt{2}$

no restrictions

	$-\infty$	-4	$-\sqrt{2}$	$\sqrt{2}$	∞
		-5	-3	0	2
$x+4$		$-$	$+$	$+$	$+$
x^2-2		$+$	$+$	$-$	$+$
x^2+4		$+$	$+$	$+$	$+$
overall		$-$	$(+)$	$-$	$(+)$

The inequality is true when $-4 \leq x \leq -\sqrt{2}$ OR $x \geq \sqrt{2}$

The inequality is true when $x \in [-4, -\sqrt{2}] \cup [\sqrt{2}, \infty)$